where $S_p^2 = \dfrac{\sum_{i=1}^{m}\left(X_i - \overline{X}\right)^2 + \sum_{j=1}^{n}\left(Y_j - \overline{Y}\right)^2}{m+n-2}$

3. If $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$ is a random sample of n paired observations drawn from a bivariate normal population, then $D_i = X_i - Y_i$, $i = 1, 2, \ldots, n$ is a random sample drawn from $N(\mu_D, \sigma_D^2)$. Here $\mu_D = \mu_X - \mu_Y$.

   Hence,

$$T_3 = \frac{\overline{D} - \mu_D}{S_D/\sqrt{n}} \sim t_{n-1}$$

2.1.2 Properties of the Student's *t*-distribution

1. *t*–distribution is symmetrical distribution with mean zero.
2. The graph of *t*-distribution is similar to normal distribution except for the following two reasons:
   i. The normal distribution curve is higher in the middle than *t*-distribution curve.
   ii. *t*–distribution has a greater spread sideways than the normal distribution curve. It means that there is more area in the tails of *t*-distribution.
3. The *t*-distribution curve is asymptotic to *X*-axis, that is, it extends to infinity on either side.
4. The shape of *t*-distribution curve varies with the degrees of freedom. The larger is the number of degrees of freedom, closeness of its shape to standard normal distribution (fig. 2.1).
5. Sampling distribution of *t* does not depend on population parameter. It depends on degrees of freedom $(n-1)$.

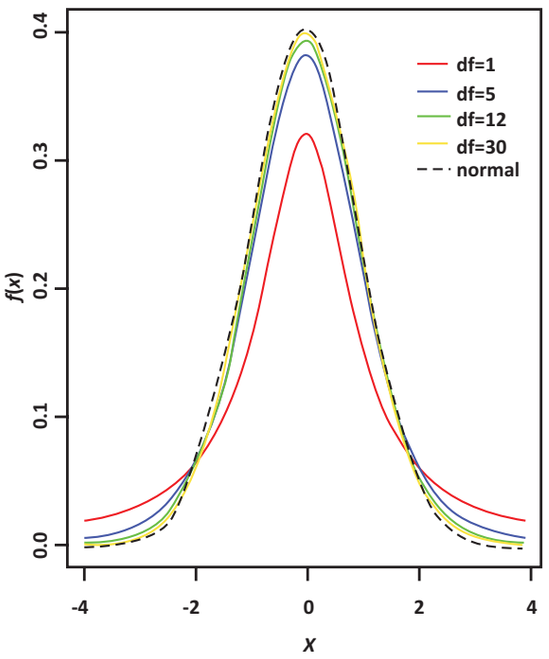


Figure 2.1. Student's *t*-distribution

2.1.3 Applications of *t*-distribution

The *t*-distribution has the following important applications in testing the hypotheses for small samples.

1. To test significance of a single population mean, when population variance is unknown, using T_1.

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